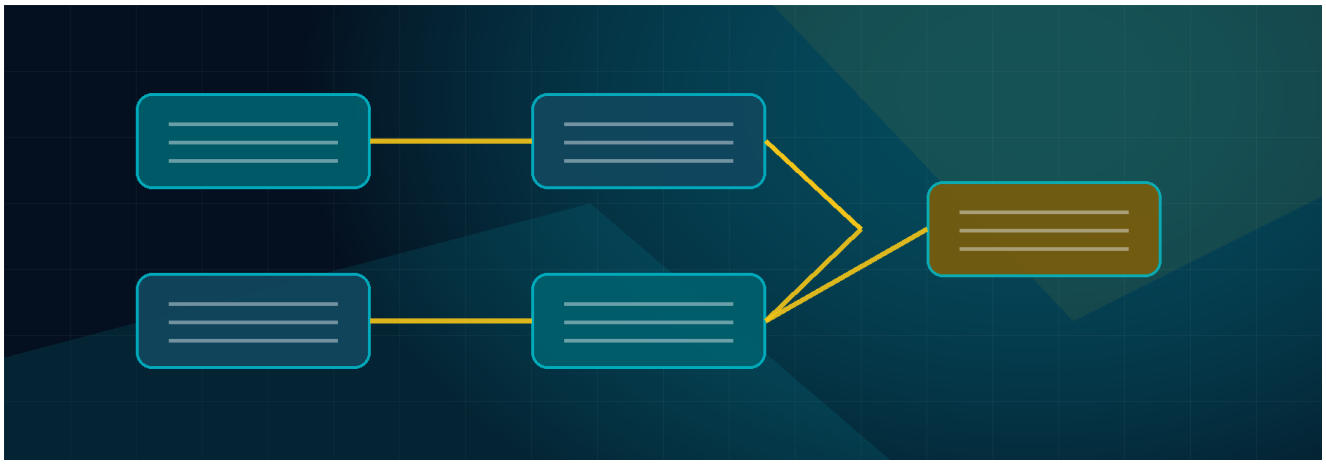
ROALLA



EXECUTIVE INDUSTRIAL SYSTEMS INTEGRATION GUIDE

ERP MES and WMS Integration Readiness Guide 2026

A decision framework for creating reliable connected information across business plant and warehouse operations

Prepared for Canadian owners, chief executives, operations, supply chain, manufacturing, warehouse, finance, technology, data, quality and transformation leaders

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Purpose

This guide helps leadership determine how ERP, planning, production, warehouse, maintenance, quality and reporting systems should work together. It connects process ownership, master data, transactions, system boundaries, interfaces, controls, analytics and support so modernization eliminates duplicate work and improves decisions instead of creating another integration layer.

EXECUTIVE DECISION DASHBOARD

Management View at a Glance

Integration succeeds when process and data ownership are explicit before interfaces are built. Management should define the authoritative system for each business object and event, simplify duplicate work, and design observable, supportable flows with reconciliation and recovery.

Executive question	ROALLA view	Decision implication
Where is duplicate work occurring?	**PREPARE**	Map re-entry, spreadsheets, manual reconciliation and conflicting reports by process.
Which system is authoritative?	**DECIDE**	Assign ownership for each master object, transaction and status before interface design.
Is real time actually required?	**CONDITIONAL**	Match latency to the operational decision and avoid unnecessary coupling.
Can failures be detected and recovered?	**PREPARE**	Instrument interfaces, queues, errors, retries, reconciliation and accountable support.
What should management authorize?	**PURSUE**	An industrial systems and data architecture diagnostic with one priority process redesign.

Recommended Management Posture

Proceed through a small number of explicit decision gates. Preserve evidence as work occurs, separate discovery from implementation, and require each investment to demonstrate value, feasibility, control effectiveness and accountable ownership before scale.

EXECUTIVE DECISION MODEL

The Eight Industrial Integration Readiness Gates

A connected industrial architecture must align the physical process, business transaction and data event. Interfaces that move inconsistent data faster increase operational risk rather than visibility.

Stage	Gate	Required conclusion	Decision evidence
01	Process	The end-to-end work is owned	Order, plan, make, move, store, ship and reconcile flows
02	System roles	Application boundaries are explicit	ERP, APS, MES, WMS, CMMS, QMS and reporting responsibilities
03	Master data	Core objects have authority	Item, BOM, routing, location, resource, customer and supplier ownership
04	Transactions	Events are timely and complete	Status, quantity, consumption, production, movement and exception rules
05	Interfaces	Integration is governed and observable	Patterns, contracts, validation, queues, retries, logs and ownership
06	Controls	Financial and operational integrity is protected	Authorization, segregation, reconciliation, traceability and audit
07	Analytics	Measures use trusted definitions	Semantic model, lineage, latency, quality and operational context
08	Operations	The architecture can be supported	Monitoring, incident, change, release, capacity, recovery and lifecycle

Decision Rule

A failed hard gate moves the initiative to Prepare or Monitor. A strong narrative, senior sponsor or attractive market cannot replace missing authority, evidence, delivery capacity or operating control.

SECTION 01

Readiness Framework

Translate ambition into a shared executive view of value, evidence, capability and risk.



Readiness Dimensions

No.	Decision area	Management evidence	Status
01	Process ownership	End-to-end flow, decisions, handoffs, exceptions and accountability	**Confirmed / Developing / Gap**
02	Application portfolio	System roles, fit, lifecycle, configuration and duplication	**Confirmed / Developing / Gap**
03	Master and transactional data	Authority, definitions, quality, timing and identifiers	**Confirmed / Developing / Gap**
04	Integration architecture	Patterns, APIs, messages, middleware, validation and observability	**Confirmed / Developing / Gap**
05	Controls and analytics	Authorization, reconciliation, traceability, lineage and metrics	**Confirmed / Developing / Gap**
06	Operating model	Support, incident, change, vendors, skills, recovery and roadmap	**Confirmed / Developing / Gap**

Evidence Standard

Assessments should distinguish verified evidence from management confidence. Where evidence is incomplete, record the gap, owner, completion date and decision that depends on it.

STRATEGIC READINESS DEPENDENCIES

What Readiness Depends On

Dependency	Why it matters	Evidence management should expect
Process	The end-to-end work is owned	Order, plan, make, move, store, ship and reconcile flows
System roles	Application boundaries are explicit	ERP, APS, MES, WMS, CMMS, QMS and reporting responsibilities
Master data	Core objects have authority	Item, BOM, routing, location, resource, customer and supplier ownership
Transactions	Events are timely and complete	Status, quantity, consumption, production, movement and exception rules
Interfaces	Integration is governed and observable	Patterns, contracts, validation, queues, retries, logs and ownership
Controls	Financial and operational integrity is protected	Authorization, segregation, reconciliation, traceability and audit
Analytics	Measures use trusted definitions	Semantic model, lineage, latency, quality and operational context
Operations	The architecture can be supported	Monitoring, incident, change, release, capacity, recovery and lifecycle

ROALLA Interpretation

Dependencies should be evaluated as a connected system. A weakness in ownership, data, architecture, delivery or governance can change the economics and risk of an otherwise attractive initiative.

SECTION 02

Detailed Readiness Assessment

Convert executive intent into verifiable requirements, accountable evidence and clear remediation actions.



READINESS CHECKLIST

Process System and Data Readiness

Completion standard Mark Ready only when the supporting evidence has been verified. Assign an evidence owner to every open item.

Gate	Readiness requirement	Ready	Gap	Evidence owner
A01	Priority order-to-ship, plan-to-produce and procure-to-receive flows have accountable owners.	☐	☐	_____
A02	Process maps show physical movement, system transactions, decisions, queues and exceptions together.	☐	☐	_____
A03	ERP, APS, MES, WMS, CMMS, QMS, controls and spreadsheet responsibilities are documented.	☐	☐	_____
A04	Duplicate capabilities, unsupported applications and operational workarounds are visible in the portfolio.	☐	☐	_____
A05	Item, BOM, routing, resource, location, customer and supplier master ownership is assigned.	☐	☐	_____
A06	Common identifiers and hierarchies reconcile across business, plant, warehouse and equipment contexts.	☐	☐	_____
A07	Data quality is measured for completeness, accuracy, uniqueness, timeliness and operational fitness.	☐	☐	_____
A08	Transaction rules define when status, quantity, consumption, production and movement become official.	☐	☐	_____
A09	Inventory, production, quality and financial reconciliation is repeatable with accountable exception handling.	☐	☐	_____
A10	Reporting definitions distinguish planned, released, started, completed, available, shipped and financially posted events.	☐	☐	_____

READINESS CHECKLIST CONTINUED

Integration Architecture and Operating Readiness

Completion standard Use Gap to create a dated remediation action and identify the management decision that cannot proceed.

Gate	Readiness requirement	Ready	Gap	Evidence owner
B01	Each interface has a business owner, technical owner, source, target, contract, frequency and service expectation.	☐	☐	_____
B02	Integration patterns match latency, volume, reliability, security and coupling requirements.	☐	☐	_____
B03	Validation prevents incomplete, duplicate, out-of-sequence or unauthorized transactions from silently posting.	☐	☐	_____
B04	Queues, retries, dead letters, logs and alerts support detection and controlled recovery.	☐	☐	_____
B05	End-to-end observability links a business transaction across applications and integration components.	☐	☐	_____
B06	Identity, service accounts, secrets, certificates and privileged changes are controlled and reviewed.	☐	☐	_____
B07	Release and change management covers coordinated testing, cutover, rollback and business validation.	☐	☐	_____
B08	Recovery priorities and procedures address middleware, interfaces, data stores and dependent operations.	☐	☐	_____
B09	Vendor contracts provide documentation, support, data access, interface rights and lifecycle visibility.	☐	☐	_____
B10	The roadmap retires duplicate work and interfaces as target capabilities are implemented.	☐	☐	_____

WEIGHTED EXECUTIVE ASSESSMENT

Readiness Scorecard

Dimension	Weight	Evidence standard	Score	Status
Process ownership	20%	End-to-end flow, decisions, handoffs, exceptions and accountability	__/20	Strong / Developing / Gap
Application portfolio	20%	System roles, fit, lifecycle, configuration and duplication	__/20	Strong / Developing / Gap
Master and transactional data	15%	Authority, definitions, quality, timing and identifiers	__/15	Strong / Developing / Gap
Integration architecture	15%	Patterns, APIs, messages, middleware, validation and observability	__/15	Strong / Developing / Gap
Controls and analytics	15%	Authorization, reconciliation, traceability, lineage and metrics	__/15	Strong / Developing / Gap
Operating model	15%	Support, incident, change, vendors, skills, recovery and roadmap	__/15	Strong / Developing / Gap
TOTAL	100%	Weighted executive assessment	__/100	Pursue / Prepare / Monitor

Decision Thresholds

Score	Posture	Management decision
75 to 100	**PURSUE**	Proceed to bounded execution subject to no hard-gate failure.
50 to 74	**PREPARE**	Complete a focused readiness plan before major commitment.
0 to 49	**MONITOR**	Resolve foundational gaps before investing further.

A single material hard-gate failure overrides the weighted score. ROALLA also applies judgment where administrative readiness is high but strategic value, adoption or commercial viability is weak.

SECTION 03

Execution and Advisory Roadmap

Start with one end-to-end process, establish authoritative data and system roles, then build reusable integration and support patterns before scaling.



Recommended 90 Day Mobilization

Timing	Focus	Key activities	Management output
Days 1 to 15	Scope	Select the priority flow, decisions, systems, owners, measures and known pain points.	Approved integration mandate
Days 16 to 35	Map	Trace process, transactions, masters, interfaces, workarounds, controls and failure modes.	Verified current-state architecture
Days 36 to 55	Design	Define target system roles, data authority, events, interfaces, controls and transition states.	Target integration blueprint
Days 56 to 75	Validate	Test data, volumes, latency, errors, security, support, cutover and business outcomes.	Validated implementation option
Days 76 to 90	Authorize	Approve roadmap, pilot, owners, governance, funding, measures and retirement plan.	Executive modernization decision

Twelve to Twenty Four Month Path

- Quarter 1: establish process, application and data ownership and resolve priority master-data defects.
- Quarter 2: implement a reusable integration pattern for one high-value flow with full observability.
- Quarters 3 and 4: extend target patterns, improve analytics and retire duplicate interfaces and workarounds.
- Year 2: optimize the application portfolio, data product ownership and industrial support model.

ROALLA ADVISORY MODEL

How ROALLA Can Help

ROALLA provides independent industrial architecture and transformation leadership. ROALLA connects operational workflows, enterprise systems, plant and warehouse technology, data governance, integration engineering, vendors and benefits so management can modernize without losing transactional or operational control.

ROALLA workstream	Executive value
Industrial systems diagnostic	Map processes, applications, data, interfaces, workarounds, controls and lifecycle risk.
Target architecture	Define ERP, planning, MES, WMS, maintenance, quality, analytics and OT boundaries.
Data ownership and quality	Establish authoritative masters, definitions, stewardship, quality measures and remediation.
Integration blueprint	Design events, APIs, messages, middleware, security, observability and recovery patterns.
Process redesign	Remove duplicate entry, reconciliation and spreadsheet dependency from priority flows.
Vendor and platform selection	Develop requirements, demonstrations, scoring, integration terms and lifecycle protections.
Migration and cutover	Plan transition states, testing, data migration, training, rollback and operational continuity.
Governance and assurance	Coordinate architecture, change, data, security, service, metrics and roadmap decisions.

Recommended Initial Mandate

- Four-week ERP MES WMS integration and data-readiness diagnostic.
- Verified process, application, master-data, interface and control fact base.
- Vendor-neutral target architecture with transition states and priority integration patterns.
- Executive roadmap and 90-day pilot plan for one end-to-end operating flow.

CONTROL FRAMEWORK

Key Risks and Controls

Risk	Exposure	ROALLA control
Interface first	Technical connections precede process ownership.	Define the business event and authority before transport.
Master conflict	Systems disagree on items, locations or structures.	Assign sources of truth and governed stewardship.
Real-time coupling	One outage stops multiple operations.	Use fit-for-purpose latency, buffering and graceful degradation.
Silent failure	Transactions fail without operational visibility.	Instrument validation, queues, alerts and reconciliation.
Spreadsheet shadow	Critical logic remains outside governed systems.	Inventory, govern and retire workarounds through process redesign.
Control gap	Automation bypasses approvals or financial integrity.	Embed authorization, segregation and reconciliation.
Vendor lock-in	Proprietary interfaces raise change cost.	Secure documentation, data rights and supported standards.
Migration overload	Big-bang change exceeds operational capacity.	Use transition states, pilots, cutover rehearsals and rollback.

Professional Advice

Qualified legal, privacy, security, tax, accounting, employment and regulatory advisors should validate matters within their professional scope. ROALLA coordinates those inputs with technology strategy, architecture, commercialization, governance and transformation decisions.

OFFICIAL EVIDENCE BASE

Selected Official Sources

These sources support the management principles summarized in this guide. Standards, laws, guidance and organizational obligations continue to evolve and must be confirmed for the specific context before acting.

NIST smart manufacturing systems design and analysis. Official source — Reference architecture, interoperability, analytics and performance-assurance considerations for smart manufacturing.

National Research Council Canada advanced manufacturing initiative. Official source — Active Canadian initiative covering digital manufacturing, sensing, AI, robotics, productivity and sustainable processes.

Canada Advanced Manufacturing Cluster. Official source — Official overview of NGen priorities including automation, machine learning, cybersecurity and SME technology adoption.

NIST Cybersecurity Framework 2.0. Official source — Outcome-based framework for governing and managing cybersecurity risk.

Canadian Centre for Cyber Security operational technology guidance. Official source — Canadian guidance for protecting operational technology used in manufacturing and other industrial environments.

Ontario Automotive Modernization Program guidelines. Official source — Sector-specific examples of eligible Industry 4.0 adoption including ERP, production tracking, predictive analytics and computer vision.

Advisory Disclaimer

This ROALLA guide is a strategic readiness and marketing document. It is not a certification, audit opinion, legal opinion, privacy opinion, security assurance, investment recommendation or guarantee of results. Management must validate current obligations, company facts and professional advice before making material decisions.